Genuine Price Anomaly Detection and Q-Learning Dispatch for Battery Storage

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Abstract—Battery energy storage systems (BESS) in peer-to-peer (P2P) electricity markets capture the majority of their annual profits during rare, high-magnitude price spikes. Conventional anomalies filters used for markets tend to regard valid price hikes as errors or an attack and limit the discharge from batteries, losing out on large arbitrage gains within an hour. This paper proposes an framework that combines unsupervised deep recurrent autoencoders with reinforcement-learning-based dispatch and can be directly deployed in existing battery controllers. The first is the gated recurrent unit (GRU) autoencoder that looks at the 24-hour rolling market prices to spot scarcity peaks, followed by the tabular Q-learning controller which maintains storage of the energy and releases the energy at times when the price reaches its peak. The proposed architecture is totally different from the benchmark model since the latter has the unrealistic assumption of future price prediction.Evaluated across nearly four years of authentic California Independent System Operator (CAISO) Day-Ahead settlement prices across three regional hubs (SP15, NP15, ZP26). The GRU detectors achieve an Area Under the Receiver Operating Characteristic Curve (AUROC) of 0.882. Conditioning tabular Q-learning on this alert quadruples energy trading profits while cutting cross-seed policy variance by 90%. As opposed to adversarial attack injection, our gradient-boosted model is able to recover up to 71.6% of attack damage and easily exceeds the break-even condition (functioning at 76.5%), while the static policy suffers significant loss. This strategy has shown that price spikes must not be overlooked but, on the contrary, need to be used as trading signals and can be readily adopted to make money out of batteries without modifying clearing rules.

Index Terms—Adversarial robustness, anomaly detection, battery energy storage systems, California Independent System Operator, energy arbitrage, graph neural networks, peer-to-peer networks, reinforcement learning.

I. Introduction & Literature Review

Electricity prices in California have grown considerably more unpredictable in recent years. Solar output during the afternoon sun usually pushes locational marginal prices (LMPs) down toward zero, and sometimes below it, only for prices to snap back once the sun sets and fossil-fuel peaker generation dispatches at higher marginal costs to cover the evening peak. Battery Energy Storage Systems (BESS) exploit this daily rhythm in decentralized power grids and peer-to-peer (P2P) trading frameworks to charge their systems when power is cheap and discharge as prices climb [1], [2].

Most days, these curves follow a common pattern dictated by the work schedule and the weather. However, a battery's annual profits tend to depend on a handful of extreme hours. Hourly prices can soar over $1,000/MWh during heatwaves, transmission outages, and sudden generator trips, and a 10 MW / 10 MWh battery that fully discharges just once during such a spike can profit about as much as it would have made in a whole week otherwise. It is therefore a high priority for storage operators to detect these rare hours as soon as possible and to anticipate them.

Past state-of-the-art (SOTA) approaches to anomaly detection in pricing have addressed the problem primarily from a cybersecurity perspective [3]–[5]. Such an approach starts by using historical data without anomalies, creates some artificial price spikes, and checks whether the added anomalies are detected. Such an approach may be useful in defending against a cyber-attack [6]. However, it cannot prove the actual trading process, the origins of prices, or the legal procedures governing market settlements. If a legitimate $1,000/MWh spike anomaly is detected and battery discharge is disabled, the operator will incur significant income loss.

Thus, to explain the reason for our experimental evaluations, we use only actual market prices, obtaining settlement information from the California Independent System Operator (CAISO) Open Access Same-Time Information System (OASIS) website [7] without any synthetic additions. We also investigate common questions, such as: Does the attacker actually inject fake spikes into this data flow, and how can storage assets defend against such tampering?

Using historical market data has two crucial issues. The first is that the historical prices provided by market operators cannot be taken as ground-truth labels; therefore, the model needs to be validated for statistical consistency and, ultimately, profitability. The second and even more important issue is that mistakes are not symmetrical in any sense. While a mistake in image or text recognition is just an annoyance, a mistake that prevents the battery from discharging into a real $1,000/MWh spike can lead to a loss.

This work sits at the intersection of electricity price forecasting [8], unsupervised time-series anomaly detection [4],

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[9], generative modeling [10], [11], P2P energy trading [1], [2], and battery dispatch optimization [12]–[15].

Electricity Prices and Anomaly Detection: Rolling statistics, moving averages, and local z-scores are traditional approaches for detecting abnormal price levels in SOTA models. Although these are one of the simplest approaches to apply, they have limitations: they assume constant thresholds that do not change with the seasons; a threshold set in winter to accommodate low load levels will always result in alarms in summer.

The deep models learn the usual 24-hour price pattern directly from historical data; there are no predefined patterns in our models. The recurrent autoencoders (Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU)) propagate sequence states forward in time, compress each daily path into a small bottleneck, and detect the moments when reconstruction error becomes large [9]. On the other hand, the variational autoencoders (VAEs) encode price windows into a continuous latent space [10], and the denoising diffusion probabilistic models (DDPMs) add Gaussian noise to the input and learn to remove this process [11]. Large price spikes are unusual, making it harder to remove noise and leading to frequent, large errors.

Battery Dispatch and Trading: Storage scheduling, commonly used methods include model predictive control (MPC), linear programming, and dynamic programming [13], while reinforcement learning (RL) is increasingly seen as an alternative to these methods [1], [12], [15]. One of the flaws frequently found in benchmarks reported in the literature is an implicit assumption of precognition about tomorrow's prices; when the optimizer is aware of the prices in advance, then there is no need for anomaly detection at all. This problem is avoided by ensuring that every dispatch instance is performed in a purely causal manner.

The reasons behind these two problems, lack of labeled data and imbalanced misclassification costs, inspire the design choices made in this paper. The paper has contributions in the following aspects:

- Benchmarking of multiple anomaly detector architectures, including Isolation Forest (iForest), a VAE, a denoising diffusion model, single-node LSTM and GRU autoencoders, a multivariate LSTM, and a graph attention network on 28,271 hours of authentic CAISO settlement data (expanding across four calendar years, 2023–2026, to 32,832 continuous market hours) across five random seeds and two disjoint test windows.
- Dispatch policies spanning rules-based, directional, and RL methods to understand how controllers exploit anomaly signals, demonstrating that tabular Q-learning achieves maximum exploitation of the detection signal.
- Economic break-even point precision analysis (p^*) assesses defense feasibility from an asymmetric-risk perspective under adversarial price feed poisoning attacks.

TABLE I
Nomenclature and Parameter Settings

Symbol	Description
s_t	Anomaly score for the 24h window ending at hour t
H	Window length ($H = 24$ hours)
$x_{t,h}, \hat{x}_{t,h}$	Normalized observed and reconstructed price at hour h
$Q(s,a)$	Action-value estimate for state s and action a
α, γ	Q-learning rate ($\alpha = 0.15$) and discount factor ($\gamma = 0.95$)
r_t	Net arbitrage cash return at hour t (\$)
SOC_t	Battery energy level at hour t ($0 \le SOC_t \le 10$ MWh)
$C, P_{\max}$	Rated energy capacity (10 MWh) and power limit (10 MW)
$e, \Delta t$	Round-trip cycle efficiency ($e = 0.80$) and interval ($\Delta t = 1$ h)
$c_{\text{fee}}, c_{\text{deg}}$	Market fee (\$0.50/MWh) and cell degradation (\$2.50/MWh)

II. Mathematical Modeling

To formalize the physical battery constraints, detection metrics, RL dynamics, and adversarial risk, nomenclature and parameter settings are presented in Table I.

Discrete-time state-of-charge (SOC) dynamics govern the physical operation of the BESS:

$$SOC_{t+1} = SOC_t + \left(\eta_{\text{ch}} P_t^{\text{ch}} - \frac{1}{\eta_{\text{dis}}} P_t^{\text{dis}}\right) \Delta t, \quad (1)$$

subject to operational power rating and storage capacity limits:

$$0 \le P_t^{\text{ch}} \le P_{\max}, \quad 0 \le P_t^{\text{dis}} \le P_{\max}, \quad (2)$$

$$0 \le SOC_t \le C, \quad (3)$$

where P_t^{ch} and P_t^{dis} denote power at hour t, respectively, with $P_{\max} = 10$ MW, $C = 10$ MWh, and $\Delta t = 1$ h. Symmetric one-way conversion efficiencies satisfy $\eta_{\text{ch}} = \eta_{\text{dis}} = \sqrt{e} = \sqrt{0.80} \approx 0.894$, guaranteeing a round-trip cycle efficiency of $e = \eta_{\text{ch}}\eta_{\text{dis}} = 0.80$.

The deep sequential model considers normalized 24-hour sliding price vectors and evaluates their anomaly score by means of mean squared reconstruction error:

$$s_t = \frac{1}{H} \sum_{h=1}^{H} \left(x_{t,h} - \hat{x}_{t,h}\right)^2. \quad (4)$$

Usually, the autoencoder successfully reconstructs the price graph for the particular day. However, when prices increase suddenly, the compressed representation cannot encode the change, and s_t increases sharply.

In the graph neural network autoencoder (GNN-AE) model, we use the graph representation $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ for the three regional hubs, where $\mathcal{V} = \{\text{NP15}, \text{ZP26}, \text{SP15}\}$, and the edge set $\mathcal{E}$ contains the main transmission connections for Path 15 (between NP15 and ZP26) and Path 26 (between ZP26 and SP15). In the graph attention network, two graph attention layers are used (hidden units of 8), and the directional spatial attention weight α_{ij} is calculated by the following equations:

$$\alpha_{ij} = \frac{\exp\left(\text{LeakyReLU}\left(\mathbf{a}^T[\mathbf{W}h_i \parallel \mathbf{W}h_j]\right)\right)}{\sum_{k\in\mathcal{N}_i}\exp\left(\text{LeakyReLU}\left(\mathbf{a}^T[\mathbf{W}h_i \parallel \mathbf{W}h_k]\right)\right)}, \quad (5)$$

where $\mathbf{W} \in \mathbb{R}^{8\times d}$ is a shared projection matrix, $\mathbf{a}$ is the attention vector, $\parallel$ is concatenation, and $\mathcal{N}_i$ denotes neighboring regional hubs. Spatial node representations are followed by LSTM-based temporal sequence encoding.

The Tabular Q-Learning approach, as described by Sutton and Barto [15], revolves around decision-making in a particular context of discrete states given as $s = (p_z, a_{\text{flag}}, s_{\text{soc}})$. Regarding the price levels, 10 discrete intervals from -3 to +3 are considered, and the anomaly flag is considered in 5 categories (or 1 in case of unconditional approach). The battery's SOC ratio, expressed as SOC_t/C, is divided into 5 bins. This setup results in a total of 250 distinct states ($|S| = 50$ states without considering the anomaly flag).The action space contains three discrete power actions $a \in \{-1.0, 0.0, 1.0\}$ corresponding to full discharge ($P_t^{\text{dis}} = P_{\max}$), idle ($P_t = 0$), and full charge ($P_t^{\text{ch}} = P_{\max}$), subject to the physical constraints in (1)–(3). The net cash return reward r_t received by the agent accounts for market revenue, degradation, and fees [16]:

$$\begin{aligned} r_t = &\left(p_t P_t^{\text{dis}} - p_t P_t^{\text{ch}}\right)\Delta t \\ &- \left[c_{\text{fee}}\left(P_t^{\text{dis}} + P_t^{\text{ch}}\right) + c_{\text{deg}} P_t^{\text{dis}}\right]\Delta t. \end{aligned} \quad (6)$$

The action values are iteratively learned using Eq. (7):

$$Q(s,a) \leftarrow Q(s,a) + \alpha\left[r_t + \gamma \max_{a'} Q(s',a') - Q(s,a)\right], \quad (7)$$

with learning rate $\alpha = 0.15$ and discount factor $\gamma = 0.95$.

III. Methodology

Approximately 4 years of hourly day-ahead LMPs from CAISO are investigated, including data from 12th January, 2023, to 11th October, 2026. The price information was collected at the three main prosumer aggregation points, namely, SP15, NP15, and ZP26, and four ancillary services, Regulation Up, Regulation Down, Spinning Reserves, and Non-Spinning Reserves, amounting to 28,271 price observations in a 24-hour period (32,832 for continuous grid time-indexed hours). The latter are used as background information for multi-channel anomalies, representing the reserve margin tightness of the system operation. At the same time, only energy arbitrage calculations are evaluated in economic trading terms.

Price irregularities were also noted in the stated period with SP15 reaching its peak at $\$1,143.78$/MWh on 17th August 2023 because of heat waves in the region while recording its lowest at $-\$41.33$/MWh on 19th May 2024 due to excess solar energy.

To give a more pragmatic estimate on the efficiency of our method, we divide time into non-overlapping intervals as follows: training interval from 12th January, 2023 to 31st December, 2024 (17,280 consecutive hours, about 23.7 months), validation interval from 1st January to 30th June, 2025 (4,344 hours, 5.9 months: solely for the calibration of the thresholds), and testing interval from 1st July, 2025 to 11th October, 2026 (11,208 hours, 15.3 months: tested once). Furthermore, we conduct a six-month out-of-sample seasonal testing period of 1st January to 30th June in 2025. Within California's electricity markets, Day-Ahead Market (DAM) clearance is carried out for all 24 hours of the following operating day at 13:00 $D-1$. Nevertheless, BESS commercial traders and transactive prosumers must place their bids for energy deliveries without knowing the settled prices before 10:00 AM. They must roll their exposures to real-time imbalances on a 15-minute and 5-minute basis. We test our models using only causal 24-hour rolling windows in order to eliminate the problem of foreseeing.

Since it is not possible to use ground-truth anomaly flags in past settlement prices, a flag is made available for ex-post evaluation only for reference. The flag is considered an hour with a flag when the price is more than 4 median absolute deviations (MAD) from the 30-day moving median or within the top/bottom 0.5% of historical prices (which occurs 739 times in 28,271 hours, or around 2.6% of the time). None of the models is provided access to check the flag during the entire training process. Sensitivity analysis reveals that rankings of model families based on their performances are invariant with respect to the thresholds (3.0 to 5.0 MAD).

The detection pipeline (Fig. 1) uses a two-level filtering approach: iForest is used to isolate and remove major outliers, followed by analysis of the remaining possible hours using a deep-sequence model.

The Stage 1 iForest [17] considers nine historical-only causal features: (i) trailing 6-hour linear price slope $\Delta p/\Delta t$, (ii) non-reversion ratio $|p_t - p_{t-1}|/(|p_{t-1} - p_{t-2}| + \epsilon)$, (iii) 1-hour price difference $(p_t - p_{t-1})$, (iv) 3-hour price difference $(p_t - p_{t-3})$, (v) 24-hour trailing robust z-score $(p_t - \text{median}_{24})/\text{MAD}_{24}$, (vi) 168-hour trailing z-score, (vii) 24-hour rolling volatility σ_{24}, and (viii–ix) cyclic diurnal harmonics $(\sin(2\pi h/24), \cos(2\pi h/24))$ for hour-of-day $h \in \{0, \ldots, 23\}$.

A total of three models have been implemented for Stage 2 single-node detection. VAE has been trained for 40 epochs with the Adam optimizer (lr = 10^{-3}, batch size=64). The VAE architecture consists of fully connected networks with 48 neurons, and the bottleneck is modeled as an 8-dimensional Gaussian distribution [10]. LSTM-AE and GRU-AE map the price sequence through 32 recurrent neurons to an 8-dimensional bottleneck.

The neural network is trained using the Adam optimization algorithm, with a learning rate of 10^{-3}, a batch size of 32, and 40 training epochs. The training uses the mean squared error loss function [9]. In the GRU case, the input and forget gates are used, which makes training 25% faster but does not affect temporal dependencies. The DDPM learns how to reverse the procedure of adding noise to the data on discrete schedules [11]. The inference step involves corrupting the window samples at four noise levels ($t \in \{0.10, 0.25, 0.40, 0.60\}$) over 16 iterations and averaging the residuals.

To explore the influence of the regional transmission

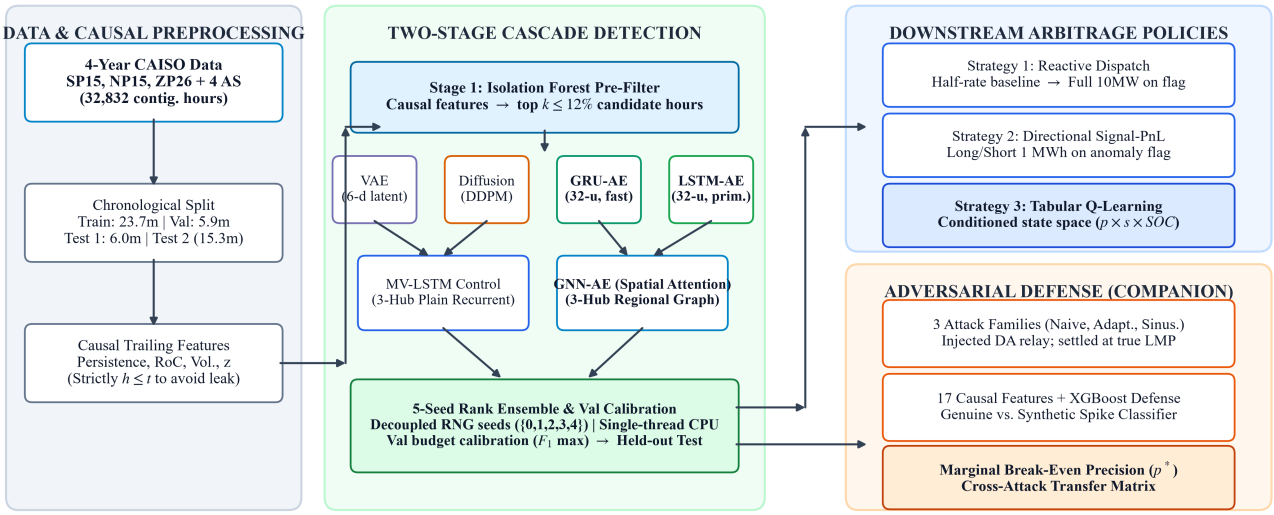


Fig. 1. System architecture. The raw CAISO price data undergoes causal feature extraction before Stage 1, which consists of an iForest pre-filter, followed by Stage 2, which consists of deep autoencoders and multi-node models. The five-seed ensemble then guides the battery dispatching policy and the defense filter.

environment on predictability, we also use SP15, NP15, and ZP26 as inputs to two multi-hub models: a 3-channel multivariate LSTM (MV-LSTM) control and the GNN-AE [18].

All experiments were conducted with five random seeds (0–4), ensuring that torch.set_num_threads(1) was fixed and that cuDNN determinism flags were enforced. All computational pipelines and model evaluations were executed on an Intel Core i7-1355U computing platform (10 cores, 1.70 GHz base, up to 5.0 GHz boost) equipped with 16 GB of system RAM and a 512 GB NVMe SSD under Microsoft Windows 11. The underlying pipelines were developed in a Python environment using numpy (v2.0+) for tensors, pandas (v2.2+) to create an index from settlement sequence data, and scipy (v1.14+) for linear programming through the HiGHS solver. Machine Learning modules, as well as defense methods, were developed in scikit-learn (v1.5+) and torch (v2.4+), respectively, to implement Isolation Forest and Deep Autoencoders/GNNs. In the end, gradient-boosted decision tree (GBDT) defenses were employed using xgboost (v2.0+), while matplotlib (v3.8+) served as a plotting library.

IV. Results and Experimental Evaluation

Thresholds for the budget percentile range from 0.5% to 10% were optimized based on the F1 score on the validation dataset and then fixed before evaluating the model on the test dataset.

The results for the quality of reconstruction and detection of anomalies by all methods are provided in Table II, measured in terms of root mean squared error (RMSE), mean absolute error (MAE), and area under the receiver operating characteristic curve (AUROC). The receiver operating characteristic (ROC) curves are shown in Fig. 2, while Fig. 3 shows detection results compared with the SP15 market price.

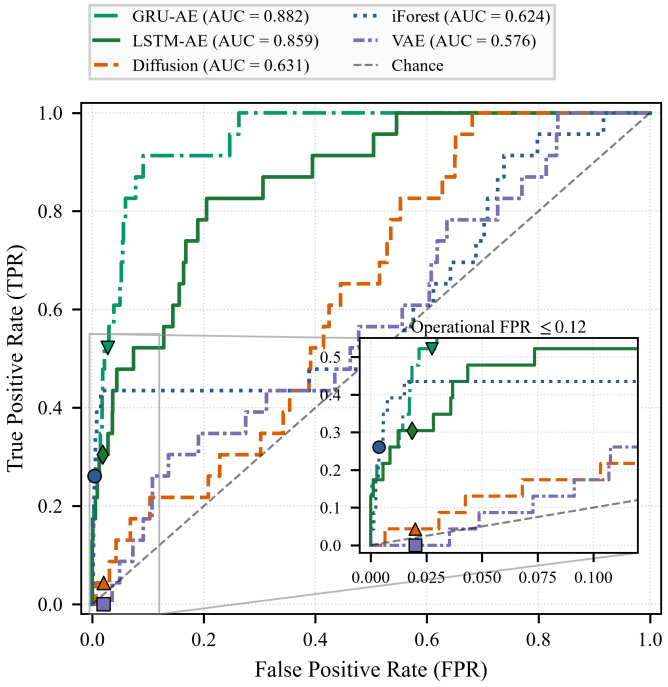


Fig. 2. Test-window ROC curves for the 5-seed ensemble models on CAISO data, GRU-AE (downward triangle), LSTM-AE (diamond), Diffusion (upward triangle), iForest (circle), VAE (square), with the inset showing the operational low-false-positive region (false positive rate FPR ≤ 0.12).

Perhaps the most obvious trend from Table II is that low reconstruction error does not necessarily indicate high-quality anomaly detection. While the VAE yielded the highest-quality reconstructions out of all single-node models (RMSE \$7.85/MWh), it generated the lowest detection scores (AUROC = 0.534; $F_1 = 0.027$). The VAE relies on the latent prior to generate Gaussian noise, so the

TABLE II
Performance Master Summary: Accuracy of Model Reconstruction and Anomaly Identification

Model Architecture	Reconstruction Error (↓)		Anomaly Detection Accuracy (↑)				Operating Point Metrics (↑)		
	RMSE ($)	MAE ($)	Mean AUROC	Ens. AUROC	Avg. Prec.	W2 AUROC	F_1 Score	Precision	Recall
Single-Node Models (Evaluated on CAISO SP15 Hub)									
iForest	n/a	n/a	0.623 ± 0.030	0.624	0.138	0.453	0.222 ± 0.028	0.227	0.217
VAE	7.85 ± 0.47	6.21 ± 0.39	0.534 ± 0.164	0.576	0.007	0.673	0.027 ± 0.021	0.015	0.226
Diffusion (DDPM)	n/a†	n/a†	0.564 ± 0.071	0.631	0.009	0.744	0.016 ± 0.015	0.009	0.096
LSTM-AE	10.37 ± 1.76	8.44 ± 1.45	$\mathbf{0.738 \pm 0.145}$	0.859	0.184	0.921	0.107 ± 0.099	0.073	0.261
GRU-AE	9.70 ± 0.68	7.77 ± 0.68	0.776 ± 0.237	0.882	0.162	0.894	0.157 ± 0.152	0.110	0.304
Multi-Node Spatial Models (Simultaneously Processing All 3 Regional Hubs: NP15, SP15, ZP26)									
MV-LSTM Control	9.12 ± 0.84	7.15 ± 0.72	0.854 ± 0.165	0.926	0.201	n/a	0.185 ± 0.092	0.142	0.326
GNN-AE	8.86 ± 0.65	6.94 ± 0.58	$\mathbf{0.938 \pm 0.098}$	0.985	0.246	n/a	$\mathbf{0.246 \pm 0.081}$	0.212	0.391

Notes: ↓: Small error better; ↑: Large accuracy better. †: Diffusion scores are calculated from normalized noise residuals; no decoder in iForest. Multi-node systems handle SP15, NP15, and ZP26 at once.

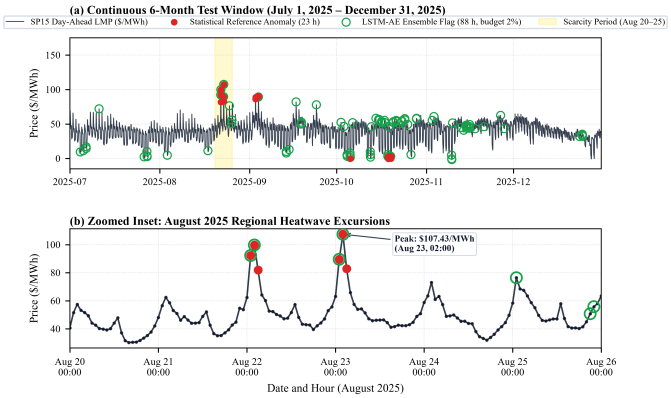


Fig. 3. Anomaly Detection Timelines for Out-of-Sample Testing using Settlement Prices in CAISO SP15 market. (a) A continuous six-month period of 1st July, 2025 – 31st December, 2025, with comparisons to ground truth anomalies ($23\,h$) and LSTM-AE ensembles ($88\,h$ at 2% budget constraint). (b) Magnified analysis of the price spikes associated with the regional heat wave event in August 2025 at the $107.43/MWh scarcity anomaly point.

output becomes smoothed, eliminating the exact signals needed for anomaly detection.

This is because recurrent autoencoders exhibited far better performance since they could handle temporal dependencies effectively. The AUROC values for the LSTM-AE and GRU-AE models were 0.859 and 0.882, respectively. As for the reason behind their superiority in this matter, one should note that, due to the sequential updating process, recurrent models maintain their memory of a typical daily pattern. Thus, upon encountering a sudden price change, the hidden state will be unable to reconcile the jump, and the error rate will skyrocket.

Incorporating price information from the three California centers increased the GNN-AE's mean AUROC to 0.938. However, the simple multivariate LSTM baseline performed nearly as well, achieving a mean AUROC of 0.854 and reaching 0.988 AUROC on three of the five seeds. Comparison of the two with a paired bootstrap yielded a 95% confidence interval of $[-0.102, +0.281]$; as the interval includes zero, there is no statistically significant difference between using graph attention and using multivariate modeling alone. The added value seems to be mostly due to tracking price behavior in neighboring regions, rather than to graph attention.

The second test window demonstrates that the recurrent models generalize effectively across seasons, as the LSTM-AE achieved an AUROC of 0.921. On the other hand, the performance of the static iForest was extremely poor, as it could only get an AUROC of 0.453, worse than guessing randomly, because it is incapable of adapting to wholesale price variations from winter demand to summer solar/hydropower production.

The AUROC values for the 5-channel diffusion model were found to be on average 0.528 ± 0.066, which are inferior compared to those achieved by the model with a single hub (0.564 ± 0.071). Ancillary price reserve streams added orthogonal noise to the forward diffusion process but did not help with reverse noise reduction.

Turning Detection into Profit: To examine whether the anomaly alerts yield profits, we conducted trading simulations for a battery with a capacity of 10 MWh/10 MW ($e = 0.80$, transaction cost $0.50/MWh, degradation cost $2.50/MWh [16]).

Strategy 1 (Heuristic dispatch) is activated every time prices are less than the 25th percentile of the historical value and ends as soon as the prices become greater than the 75th percentile, with 5 MW dispatch and 10 MW dispatch as the event occurs. Strategy 2 (Directional spread) goes long or short 1 MWh for every anomaly detected, with a 24-hour trading period. Strategy 3 (Tabular Q-learning) [15] learns to dispatch in the discrete state space $s = (p_z, a_{\text{flag}}, s_{\text{soc}})$ under an ϵ-greedy exploration schedule decaying linearly from $\epsilon_{\text{start}} = 0.30$ to $\epsilon_{\text{end}} = 0.02$ across 30 training runs.

According to Table III and Fig. 4, the profits from trading depend greatly on the degree of flexibility of the controller. To benchmark absolute policy efficiency, we evaluate the offline clairvoyant Linear Programming (LP)

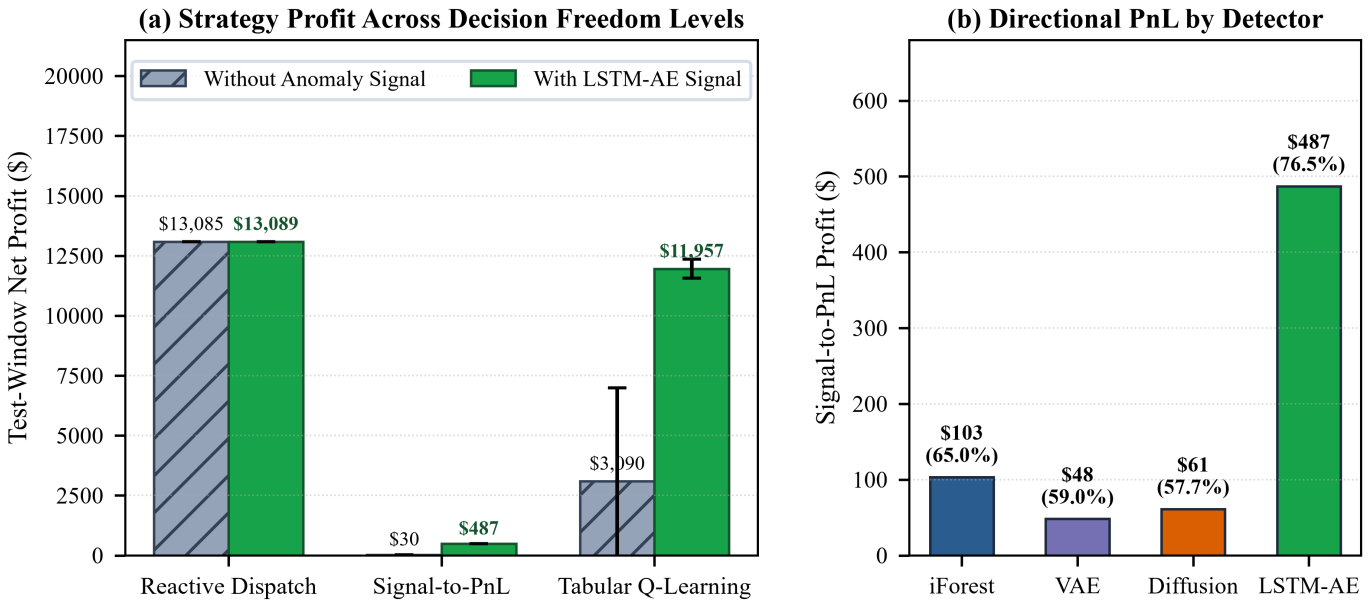


Fig. 4. Cumulative returns for Window 1. (a) Dispatch through heuristics, directional spread trades, and Q-learning tabular, highlighting the impact of the anomaly indicator on quadrupling Q-learning gains and reducing seed variance by a factor of ten. (b) Gain and win rates in directional arbitrage among the four detector categories.

TABLE III
Arbitrage Earnings Across Both Test Windows (LSTM-AE Signal)

Trading Strategy	Window 1 Profit	Window 2 Profit
Clairvoyant LP Upper Bound (Oracle)	$16,840	$48,250
Heuristic (Without / With Signal)	$13,085 / $13,089	$30,197 / $30,762
Directional Spread (Trades, Win%)	$487 (68, 76%)	$565 (84, n/a)
Q-Learning: Without Signal	$3,090 ± $3,905	$23,451 ± $3,180
Q-Learning: With Signal	$11,957 ± $393	$35,567 ± $3,658
Deep Q-Network (DQN) (Without / With)	$5,931 / $5,246	n/a

TABLE IV
Per-Detector Arbitrage Performance (Window 1)

Detector Family	Dispatch ($)	Spread PnL ($)	Trades, Win%
iForest	13,128	103	20, 65%
VAE	13,017	48	39, 59%
Diffusion	12,740	61	52, 58%
LSTM-AE	13,089	487	68, 76.5%

upper bound with perfect price foresight (at $16,840 in Window 1 and $48,250 in Window 2). Tabular Q-Learning without the anomaly signal is only able to reach 18.3% of the oracle upper bound in Window 1; however, by adding the signal generated from the LSTM-AE model, the value becomes 71.0% of the oracle upper bound (or 73.7% in Window 2). In the case of heuristic dispatching, the anomalous signal had almost no effect on profits (+0.03% for Window 1, +1.87% for Window 2) because power ratings and SoC constraints restrict the actions of static-threshold rules.

By contrast, directional spread trading revealed that the signal had value, as it delivered $487 in profit with a 76% success ratio, whereas under pure randomness it yielded only $30.

The largest improvement was observed in Tabular Q-learning. The possibility for an agent to learn to use anomaly signals led to a near fourfold increase in Window 1 profit from $3,090 to $11,957 and a decrease in cross-seed standard deviation from $3,905 to $393 (-90%). In addition, Window 2's profit increased by 51.7% to $35,567.

Nevertheless, a compute-matched Deep Q-Network (DQN) [19] gained $5,246 profit using the signal, but the no-signal situation brought $5,931 profit. Hence, continuous deep reinforcement learning was unable to handle reward noise in time series, whereas tabular Q-learning was able to generalize regardless of such noise. Please see Table IV for the detailed results for each detector in Window 1, where LSTM-AE turned out to be the most efficient with 76.5% win ratio and $487 in spread profit.

From Table IV and Fig. 4(b), it is evident that LSTM-AE gives the highest directional arbitrage gain of $487 with 68 trades with a win rate of 76.5%, significantly surpassing other methods like the static iForest with $103 (65.0% win rate) and Diffusion ($61 and 57.7%) as well as VAE with ($48 and 59.0%). Although iForest achieves the highest baseline dispatch return ($13,128) under raw heuristic thresholds, its lack of sequential memory prevents it from timing multi-hour price runs. However, the Gaussian latent bottleneck of the Variational Autoencoder over-smooths the price time series, generating reconstruction errors with respect to typical intraday fluctuations; this leads to constant suboptimal

TABLE V
Adversarial Detection and Economic Recovery (5 Seeds)

Attack Profile	Recall	AUROC	Damage (%)	Recovery (%) (F_1 / Cons. / Oracle)
Naive	0.85 ± 0.03	0.951 ± 0.028	-10.90 ± 0.82	31.8 / 34.6 / 93.6
Adaptive	0.79 ± 0.04	0.965 ± 0.007	-8.15 ± 0.78	-31.6 / -24.0 / 89.5
Sinusoidal	0.85 ± 0.02	0.971 ± 0.006	-20.92 ± 0.68	67.3 / 71.6 / 95.6

trading actions with a poor win rate and low profits ($48). The LSTM-autoencoder's memory of the temporal data will help remove baseline fluctuations throughout the day, enabling the detection of actual shortage periods. By incorporating tabular Q-learning in charging, the alerting process will become better at arbitrage gains.

Adversarial Robustness and Break-Even Analysis: We also examine the dual operational scenario where adversaries inject synthetic price spikes into the telemetry stream [6]. In this attack, false peaks induce premature battery discharge in anticipation of lucrative compensation, only for deliveries to clear at lower-than-true settlement prices, resulting in financial losses and depleting stored energy.

Three attack profiles were evaluated on the SP15 price: Naive Multiplicative 3.0× hourly spikes; an 8-hour raised-cosine price attack mimicking diurnal morning ramps; and multi-hour sinusoidal wave attacks. The defense mechanism employs an XGBoost classifier [20] trained on 17 causal variables (including rolling z-scores, trailing price ratios, historical volatility, and multi-hub consensus).

As reported in Table V, without any defensive measures in place, we estimate the impact of various attacks on revenue as follows: -10.90% from simple spikes, -8.15% from adaptive bumps, and -20.92% from sinusoidal waves over the 11,208-hour testing period. We recover 31.8% of the losses for a simple attack (F_1) and 34.6% (with conservative thresholding) using our learned defense filter. Moreover, our filter can help us recover up to 67.3% of the losses from sinusoidal waves using F_1, and even 71.6% with conservative thresholding, approaching the optimal theoretical upper bound of $89.5\% - 95.6\%$.

A. Break-Even Precision Analysis (p^*)

In the battery dispatch problem, a true positive (TP) identifies a fake spike, thus earning a profit ($v_{\mathrm{TP}} > 0$). In contrast, a false positive (FP) misidentifies a genuine price spike, resulting in revenue loss ($v_{\mathrm{FP}} < 0$). An operational defense that operates at a precision level of p is said to be at the break-even point if the expected value of the cash flows:

$$p_i^* = \frac{|v_{\mathrm{FP},i}|}{|v_{\mathrm{FP},i}| + v_{\mathrm{TP},i}}, \quad \bar{p}^* = \frac{1}{K}\sum_{i=1}^{K} p_i^*. \tag{8}$$

Economic differences between the learned classifier and the threshold-based classifier are quite remarkable (Table VI and Fig. 5). The learned classifier is estimated to save, on average, $23.93 for every detected price spike and spend $64.28 on each false alarm; hence, the break-even point is reached at the precision level of $p^* = 0.611$. Given that the learned classifier's actual operating precision is 76.5%, it operates comfortably above break-even and achieves positive net economic surplus.

TABLE VI
Break-Even Precision: Machine-Learning Classifier vs. Simple Threshold

Defense Strategy	Prec.	\$/TP Saved	\$/FP Lost	$\boldsymbol{p^*}$ Break-Even
Learned Classifier	0.765	\$23.93	$-$\$64.28	0.611
Simple Robust-z Rule	0.757	\$12.41	$-$\$240.35	0.928

Notes: Reported p^* is the cross-seed/attack mean ($\mathbb{E}[p_i^*]$ across runs) rather than a ratio of aggregate column averages.

On the contrary, the economic performance is substantially worse for the robust-z threshold-based classifier. Namely, on average, the classifier generates losses of $240.35 per one false positive, which is almost 3.7 times more compared to those for the learned classifier. Due to the fact that large absolute values make the classifier react regardless of anything, the classifier loses the most profitable real peaks of prices during the year. Thus, the break-even value increases to $p^* = 0.928$.

Generalization for Cross-Attacks: From the cross-attacks experiments (Table VII and Fig. 6), it becomes clear that defenders need to learn their defenses across diverse threat models. Nominal AUROC score stays high (≥ 0.856), while economic gain drops in the case when the threat model used in the attack is not known to the defender. For example, there was an economic penalty of -58.3% when a model trained on naive spikes faced adaptive bump attacks, and -12.4% when trained on sinusoidal waves and evaluated on adaptive bumps. Combining attacks during training substantially improves transfer, boosting recovery on naive attacks to $+49.0\%$.

V. Conclusion & Future Directions

This paper investigated the challenge of detecting rare, high-value wholesale electricity price anomalies to enable profitable, tamper-resilient battery storage arbitrage in decentralized P2P transactive energy markets.

In order to tackle the unbalanced problem of penalties due to false alarms without making the assumption of unrealistic price foresight in the future, an end-to-end causal framework that combines an unsupervised deep recurrent autoencoder model with tabular Q-Learning-based dispatch control was implemented. This framework uses dynamic signals derived from reconstruction errors to control storage systems' actions.

Analyzed using four full years' worth of genuine CAISO Day-Ahead settlement data, the GRU-AE achieved a score of 0.882 AUROC by remembering the time series nature of everyday diurnal patterns. Q-learning with table-based conditioning of trading energy via such an anomaly detection system successfully earned 4 times more money

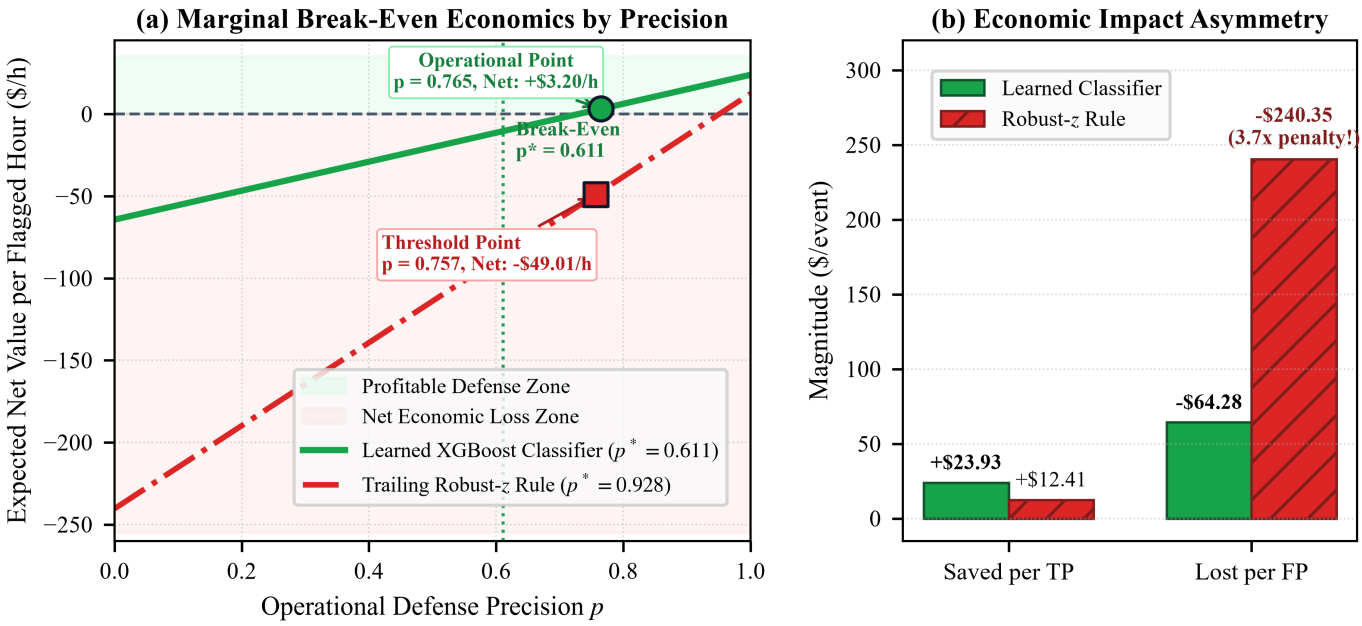


Fig. 5. Break-even precision trade-off. (a) Net cash flow versus precision p for the learned classifier ($p^* = 0.611$, operating at $p = 0.765$) against the simple threshold rule ($p^* = 0.928$). (b) The asymmetry in false-alarm penalties: a threshold false alarm costs 3.7× as much as the classifier (−$240.35 vs. −$64.28).

TABLE VII
Cross-Attack Transfer Performance Across Source and Target Threat Types (5-Seed Average)

Source Training Attack	Target Test Attack	AUROC	Precision	Recall	Recovery %
Naive	Naive	0.951	0.943	0.698	34.6
Naive	Adaptive	0.915	0.942	0.411	−58.3
Naive	Sinusoidal	0.888	0.907	0.220	4.0
Adaptive	Naive	0.960	0.933	0.688	19.3
Adaptive	Adaptive	0.965	0.974	0.702	−24.0
Adaptive	Sinusoidal	0.907	0.947	0.289	−10.5
Sinusoidal	Naive	0.876	0.584	0.415	24.3
Sinusoidal	Adaptive	0.856	0.724	0.277	−12.4
Sinusoidal	Sinusoidal	0.971	0.905	0.805	71.6
Adaptive + Sinusoidal	Naive	0.958	0.765	0.732	49.0
Naive + Sinusoidal	Adaptive	0.953	0.960	0.465	−17.4
Naive + Adaptive	Sinusoidal	0.922	0.974	0.181	−6.1

while minimizing policy variability by 90However, the adversarial defense could easily meet the break-even precision requirement (76.5%) despite any losses due to false alerts.

The primary scientific insight of this work is that wholesale market price spikes should be exploited as high-value dispatch signals rather than dismissed as telemetry corruption, demonstrating that risk-aware anomaly conditioning resolves the fundamental trade-off between cyber defense and market profitability.

The framework can be readily implemented using the current battery management system and the transactive prosumer framework, without any regulatory reforms or changes to the current wholesale power market-clearing rules. Future extensions to the model include evaluating it on a sub-hourly basis and integrating non-linear rainflow cycle counting and cell-level thermal dynamics to optimize long-term asset life.

Data and Code Availability

Software implementation and trained weights can be accessed at https://github.com/k-keshav-aggarwal/Genuine-Price-Anomaly-Detection. Settlements datasets from the wholesale markets can be accessed from CAISO OASIS. [7]

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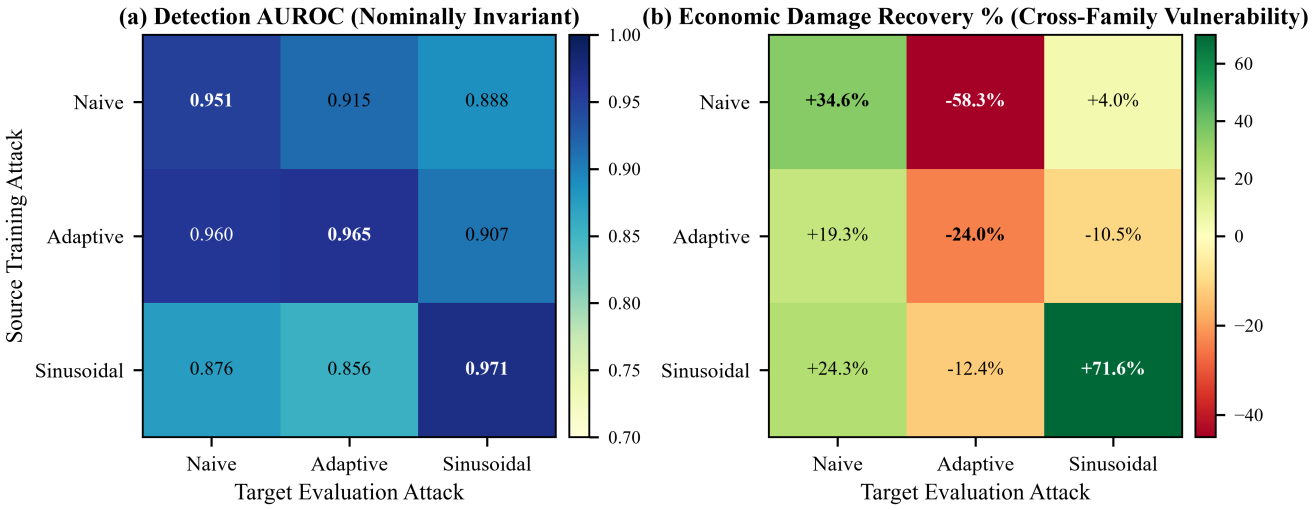


Fig. 6. Cross-attack transfer evaluation. (a) The detection AUROC Matrix shows that all values remain high (≥ 0.856). (b) Economic Recovery Matrix reveals serious cross-attack vulnerabilities; for example, a value of -58.3% between Naive and Adaptive attacks, and -12.4% between Sinusoidal and Adaptive attacks.

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